

CHAPTER-14

LIMITS

1. Introduction to Limits

What is a Limit?

A **limit** describes the value that a function approaches as the input approaches a certain number.

$$\lim_{x \rightarrow a} f(x) = L$$

This means:

As x gets closer and closer to a , the function $f(x)$ gets closer and closer to L .

Definition of Limit

Understanding Limit

A limit tells us the value that a function approaches as x gets closer and closer to a particular number.

Consider the function:

$$f(x) = x + 5$$

We want to study what happens when x approaches 2.

Approaching 2 from the Left Side

Take values slightly smaller than 2:

1.9, 1.99, 1.999

x	$f(x) = x + 5$
1.9	6.9
1.99	6.99
1.999	6.999

We can see that as x gets closer to 2 from the left side,

$$f(x)$$

gets closer to 7.

Therefore,

$$\lim_{x \rightarrow 2^-} (x + 5) = 7$$

Here,

$$2^-$$

means approaching 2 from values less than 2.

Approaching 2 from the Right Side

Now take values slightly greater than 2:

2.1, 2.01, 2.001

x	$f(x) = x + 5$
2.1	7.1
2.01	7.01
2.001	7.001

Again, we observe that as x gets closer to 2 from the right side,

$$f(x)$$

approaches 7.

Hence,

$$\lim_{x \rightarrow 2^+} (x + 5) = 7$$

Here,

$$2^+$$

means approaching 2 from values greater than 2.

Final Conclusion

Since both left-hand limit and right-hand limit are equal to 7,

$$\lim_{x \rightarrow 2^-} (x + 5) = \lim_{x \rightarrow 2^+} (x + 5) = 7$$

therefore,

$$\lim_{x \rightarrow 2} (x + 5) = 7$$

Important Idea

The function does not need to actually reach the value first.
A limit only studies what value the function is approaching.

Simple Idea

Suppose:

$$f(x) = 2x + 1$$

Find:

$$\lim_{x \rightarrow 2} (2x + 1)$$

Substitute $x = 2$:

$$2(2) + 1 = 5$$

Therefore,

$$\boxed{\lim_{x \rightarrow 2} (2x + 1) = 5}$$

2. Why Do We Need Limits?

Importance of Limits

Limits are used to:

- Study behavior near a point
- Understand continuity
- Define derivatives and integrals
- Handle functions that are undefined at some points

3. Left-Handed Limit

Definition

The **left-handed limit** means approaching a number from the **left side**.

Notation:

$$\lim_{x \rightarrow a^-} f(x)$$

The symbol a^- means:

$$x < a$$

and x moves toward a .

Example 1

Consider:

$$f(x) = x + 3$$

Find:

$$\lim_{x \rightarrow 2^-} (x + 3)$$

Take values slightly less than 2:

$$1.9, 1.99, 1.999$$

Then:

$$f(1.9) = 4.9$$

$$f(1.99) = 4.99$$

$$f(1.999) = 4.999$$

The values approach 5.

Hence,

$$\boxed{\lim_{x \rightarrow 2^-} (x + 3) = 5}$$

4. Right-Handed Limit

Definition

The **right-handed limit** means approaching a number from the **right side**.

Notation:

$$\lim_{x \rightarrow a^+} f(x)$$

The symbol a^+ means:

$$x > a$$

and x moves toward a .

Example 2

Find:

$$\lim_{x \rightarrow 2^+} (x + 3)$$

Take values slightly greater than 2:

$$2.1, 2.01, 2.001$$

Then:

$$f(2.1) = 5.1$$

$$f(2.01) = 5.01$$

$$f(2.001) = 5.001$$

The values approach 5.

Therefore,

$$\lim_{x \rightarrow 2^+} (x + 3) = 5$$

5. When Does a Limit Exist?

Condition for Existence of Limit

A limit exists only when:

$$\lim_{x \rightarrow a^-} f(x) = \lim_{x \rightarrow a^+} f(x)$$

That means:

- Left-hand limit = Right-hand limit

6. Example Where Limit Exists

Consider:

$$f(x) = x^2$$

Find:

$$\lim_{x \rightarrow 3} x^2$$

Left-hand limit:

$$\lim_{x \rightarrow 3^-} x^2 = 9$$

Right-hand limit:

$$\lim_{x \rightarrow 3^+} x^2 = 9$$

Since both are equal,

$$\lim_{x \rightarrow 3} x^2 = 9$$

7. Example Where Limit Does NOT Exist

Define:

$$f(x) = \begin{cases} 1, & x < 0 \\ 3, & x > 0 \end{cases}$$

Find:

$$\lim_{x \rightarrow 0} f(x)$$

Left-hand limit:

$$\lim_{x \rightarrow 0^-} f(x) = 1$$

Right-hand limit:

$$\lim_{x \rightarrow 0^+} f(x) = 3$$

Since:

$$1 \neq 3$$

the limit does not exist.

$\lim_{x \rightarrow 0} f(x)$ does not exist
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8. Quick Summary

Summary Table

Concept	Meaning
$\lim_{x \rightarrow a} f(x)$	General limit
$\lim_{x \rightarrow a^-} f(x)$	Approach from left side
$\lim_{x \rightarrow a^+} f(x)$	Approach from right side
Limit exists	LHL = RHL

9. Practice Questions

1. Find:

$$\lim_{x \rightarrow 4} (2x - 1)$$

2. Find:

$$\lim_{x \rightarrow 1^-} (x^2 + 2)$$

3. Find:

$$\lim_{x \rightarrow 1^+} (x^2 + 2)$$

4. Determine whether the limit exists:

$$f(x) = \begin{cases} 2, & x < 5 \\ 7, & x > 5 \end{cases}$$

Algebra of Limits

Introduction

Algebra of limits helps us evaluate limits easily using simple algebraic operations. If

$$\lim_{x \rightarrow a} f(x) = L$$

and

$$\lim_{x \rightarrow a} g(x) = M$$

then we can apply different algebraic rules on limits.

1. Sum Rule of Limits

Sum Rule

The limit of a sum is equal to the sum of the limits.

$$\lim_{x \rightarrow a} [f(x) + g(x)] = \lim_{x \rightarrow a} f(x) + \lim_{x \rightarrow a} g(x)$$

Example

Find:

$$\lim_{x \rightarrow 2} (x^2 + 3x)$$

Using limit laws:

$$= \lim_{x \rightarrow 2} x^2 + \lim_{x \rightarrow 2} 3x$$

Now substitute $x = 2$:

$$= 2^2 + 3(2)$$

$$= 4 + 6$$

$$= 10$$

Therefore,

$$\lim_{x \rightarrow 2} (x^2 + 3x) = 10$$

2. Difference Rule of Limits

Difference Rule

The limit of a difference is equal to the difference of the limits.

$$\lim_{x \rightarrow a} [f(x) - g(x)] = \lim_{x \rightarrow a} f(x) - \lim_{x \rightarrow a} g(x)$$

Example

Find:

$$\lim_{x \rightarrow 3} (x^2 - 2x)$$

Substitute $x = 3$:

$$= 3^2 - 2(3)$$

$$= 9 - 6$$

$$= 3$$

Hence,

$$\lim_{x \rightarrow 3} (x^2 - 2x) = 3$$

3. Constant Multiple Rule

Constant Multiple Rule

A constant can be taken outside the limit.

$$\lim_{x \rightarrow a} [c f(x)] = c \lim_{x \rightarrow a} f(x)$$

Example

Find:

$$\lim_{x \rightarrow 2} (5x)$$

Take constant 5 outside:

$$= 5 \lim_{x \rightarrow 2} x$$

Now substitute $x = 2$:

$$= 5(2)$$

$$= 10$$

Therefore,

$$\boxed{\lim_{x \rightarrow 2} (5x) = 10}$$

4. Product Rule of Limits

Product Rule

The limit of a product is equal to the product of the limits.

$$\lim_{x \rightarrow a} [f(x)g(x)] = \left(\lim_{x \rightarrow a} f(x) \right) \left(\lim_{x \rightarrow a} g(x) \right)$$

Example

Find:

$$\lim_{x \rightarrow 2} (x)(x + 1)$$

Substitute $x = 2$:

$$= (2)(2 + 1)$$

$$= (2)(3)$$

$$= 6$$

Hence,

$$\boxed{\lim_{x \rightarrow 2} x(x + 1) = 6}$$

5. Quotient Rule of Limits

Quotient Rule

The limit of a quotient is equal to the quotient of the limits.

$$\lim_{x \rightarrow a} \frac{f(x)}{g(x)} = \frac{\lim_{x \rightarrow a} f(x)}{\lim_{x \rightarrow a} g(x)}$$

provided that

$$\lim_{x \rightarrow a} g(x) \neq 0$$

Example

Find:

$$\lim_{x \rightarrow 2} \frac{x+3}{x+1}$$

Substitute $x = 2$:

$$\begin{aligned} &= \frac{2+3}{2+1} \\ &= \frac{5}{3} \end{aligned}$$

Therefore,

$$\boxed{\lim_{x \rightarrow 2} \frac{x+3}{x+1} = \frac{5}{3}}$$

6. Power Rule of Limits

Power Rule

If the limit of a function exists, then:

$$\lim_{x \rightarrow a} [f(x)]^n = \left(\lim_{x \rightarrow a} f(x) \right)^n$$

Example

Find:

$$\lim_{x \rightarrow 2} (x^3)$$

Substitute $x = 2$:

$$= 2^3$$

$$= 8$$

Hence,

$$\boxed{\lim_{x \rightarrow 2} x^3 = 8}$$

7. Root Rule of Limits

Root Rule

$$\lim_{x \rightarrow a} \sqrt[n]{f(x)} = \sqrt[n]{\lim_{x \rightarrow a} f(x)}$$

Example

Find:

$$\lim_{x \rightarrow 4} \sqrt{x}$$

Substitute $x = 4$:

$$= \sqrt{4}$$

$$= 2$$

Therefore,

$$\boxed{\lim_{x \rightarrow 4} \sqrt{x} = 2}$$

8. Summary Table

Rule	Formula
Sum Rule	$\lim(f + g) = \lim f + \lim g$
Difference Rule	$\lim(f - g) = \lim f - \lim g$
Constant Rule	$\lim(cf) = c \lim f$
Product Rule	$\lim(fg) = (\lim f)(\lim g)$
Quotient Rule	$\lim \frac{f}{g} = \frac{\lim f}{\lim g}$
Power Rule	$\lim(f^n) = (\lim f)^n$
Root Rule	$\lim \sqrt[n]{f(x)} = \sqrt[n]{\lim f(x)}$

9. Practice Questions

1. Find:

$$\lim_{x \rightarrow 1} (x^2 + 4x)$$

2. Find:

$$\lim_{x \rightarrow 2} (3x^2 - 5)$$

3. Find:

$$\lim_{x \rightarrow 3} \frac{x + 1}{x - 1}$$

4. Find:

$$\lim_{x \rightarrow 4} \sqrt{x + 5}$$

5. Find:

$$\lim_{x \rightarrow 2} (x^2)(x + 3)$$

Important Note

Algebra of limits allows us to evaluate complicated limits by breaking them into smaller and easier parts.

Limits of Polynomial and Rational Functions

Introduction

Polynomial and rational functions are among the easiest functions for evaluating limits.

For these functions, we usually find the limit by directly substituting the value of x .

1. Limits of Polynomial Functions

Polynomial Limit Theorem

If $p(x)$ is a polynomial function, then

$$\lim_{x \rightarrow a} p(x) = p(a)$$

This means we can directly substitute $x = a$.

Example 1 : Linear Polynomial

Evaluate:

$$\lim_{x \rightarrow 4} (3x + 2)$$

Step 1: Substitute $x = 4$

$$= 3(4) + 2$$

$$= 12 + 2$$

$$= 14$$

Therefore,

$$\boxed{\lim_{x \rightarrow 4} (3x + 2) = 14}$$

Example 2 : Quadratic Polynomial

Evaluate:

$$\lim_{x \rightarrow 2} (x^2 + 5x - 1)$$

Substitute $x = 2$:

$$= 2^2 + 5(2) - 1$$

$$= 4 + 10 - 1$$

$$= 13$$

Hence,

$$\boxed{\lim_{x \rightarrow 2} (x^2 + 5x - 1) = 13}$$

Example 3 : Cubic Polynomial

Evaluate:

$$\lim_{x \rightarrow -1} (2x^3 - x + 4)$$

Substitute $x = -1$:

$$= 2(-1)^3 - (-1) + 4$$

$$= -2 + 1 + 4$$

$$= 3$$

Therefore,

$$\boxed{\lim_{x \rightarrow -1} (2x^3 - x + 4) = 3}$$

2. Limits of Rational Functions

Rational Function Limit Theorem

If

$$r(x) = \frac{p(x)}{q(x)}$$

is a rational function and

$$q(a) \neq 0$$

then

$$\lim_{x \rightarrow a} r(x) = r(a)$$

That means we substitute directly if the denominator is not zero.

Example 4 : Simple Rational Function

Evaluate:

$$\lim_{x \rightarrow 3} \frac{x + 4}{x + 1}$$

Substitute $x = 3$:

$$= \frac{3 + 4}{3 + 1}$$

$$= \frac{7}{4}$$

Therefore,

$$\boxed{\lim_{x \rightarrow 3} \frac{x + 4}{x + 1} = \frac{7}{4}}$$

Example 5 : Rational Function with Quadratic

Evaluate:

$$\lim_{x \rightarrow 2} \frac{x^2 + 3x + 1}{x + 5}$$

Substitute $x = 2$:

$$\begin{aligned} &= \frac{2^2 + 3(2) + 1}{2 + 5} \\ &= \frac{4 + 6 + 1}{7} \\ &= \frac{11}{7} \end{aligned}$$

Hence,

$$\boxed{\lim_{x \rightarrow 2} \frac{x^2 + 3x + 1}{x + 5} = \frac{11}{7}}$$

3. Indeterminate Form $\frac{0}{0}$

Important Concept

If direct substitution gives

$$\frac{0}{0}$$

then the limit cannot be evaluated immediately.

We simplify the expression first.

Example 6 : Factorization Method

Evaluate:

$$\lim_{x \rightarrow 5} \frac{x^2 - 25}{x - 5}$$

Step 1: Direct substitution

$$= \frac{5^2 - 25}{5 - 5} = \frac{0}{0}$$

This is an indeterminate form.

Step 2: Factorize numerator

$$x^2 - 25 = (x - 5)(x + 5)$$

So,

$$\frac{x^2 - 25}{x - 5} = \frac{(x - 5)(x + 5)}{x - 5}$$

Cancel $x - 5$:

$$= x + 5$$

Step 3: Substitute $x = 5$

$$= 5 + 5$$

$$= 10$$

Therefore,

$$\boxed{\lim_{x \rightarrow 5} \frac{x^2 - 25}{x - 5} = 10}$$

Example 7 : Higher Polynomial

Evaluate:

$$\lim_{x \rightarrow 1} \frac{x^3 - 1}{x - 1}$$

Step 1: Factorize

$$x^3 - 1 = (x - 1)(x^2 + x + 1)$$

Thus,

$$\frac{x^3 - 1}{x - 1} = x^2 + x + 1$$

Step 2: Substitute $x = 1$

$$= 1 + 1 + 1$$

$$= 3$$

Hence,

$$\boxed{\lim_{x \rightarrow 1} \frac{x^3 - 1}{x - 1} = 3}$$

4. Important Theorem

Theorem

For any positive integer n ,

$$\lim_{x \rightarrow a} \frac{x^n - a^n}{x - a} = na^{n-1}$$

Example 8

Evaluate:

$$\lim_{x \rightarrow 2} \frac{x^4 - 16}{x - 2}$$

Using the theorem:

$$n = 4, \quad a = 2$$

Therefore,

$$= 4(2)^{4-1}$$

$$= 4(8)$$

$$= 32$$

Hence,

$$\boxed{32}$$

5. Summary Table

Function Type	Method
Polynomial Function	Direct substitution
Rational Function	Substitute if denominator $\neq 0$
Indeterminate Form $\frac{0}{0}$	Factorize and simplify
Expression of form $\frac{x^n - a^n}{x - a}$	Use theorem

6. Practice Questions

1. Evaluate:

$$\lim_{x \rightarrow 2} (4x^2 - 3x + 7)$$

2. Find:

$$\lim_{x \rightarrow -1} (x^3 + 2x^2 - 5)$$

3. Evaluate:

$$\lim_{x \rightarrow 4} \frac{2x + 1}{x - 2}$$

4. Find:

$$\lim_{x \rightarrow 3} \frac{x^2 - 9}{x - 3}$$

5. Evaluate:

$$\lim_{x \rightarrow 2} \frac{x^3 - 8}{x - 2}$$

6. Find:

$$\lim_{x \rightarrow 1} \frac{x^4 - 1}{x - 1}$$

7. Evaluate:

$$\lim_{x \rightarrow -2} \frac{x^2 + 5x + 6}{x + 2}$$

8. Find:

$$\lim_{x \rightarrow 5} \frac{x^2 - 10x + 25}{x - 5}$$

Key Idea

Most polynomial and rational limits can be solved using:

- Direct substitution
- Factorization
- Algebraic simplification

Limits of Trigonometric Functions

Important Trigonometric Limits

The following limits are very important in calculus:

$$\lim_{x \rightarrow 0} \frac{\sin x}{x} = 1$$

and

$$\lim_{x \rightarrow 0} \frac{\tan x}{x} = 1$$

These formulas are valid only when angles are measured in radians.

1. Limit of Type $\frac{\sin ax}{bx}$

Formula

If a and b are constants and $b \neq 0$, then

$$\lim_{x \rightarrow 0} \frac{\sin(ax)}{bx} = \frac{a}{b}$$

Example 1

Evaluate:

$$\lim_{x \rightarrow 0} \frac{\sin(5x)}{2x}$$

Step 1: Rewrite the expression

$$= \frac{5}{2} \cdot \frac{\sin(5x)}{5x}$$

Step 2: Apply standard limit

$$\lim_{x \rightarrow 0} \frac{\sin(5x)}{5x} = 1$$

Therefore,

$$\begin{aligned} &= \frac{5}{2} \times 1 \\ &= \frac{5}{2} \end{aligned}$$

Hence,

$$\boxed{\lim_{x \rightarrow 0} \frac{\sin(5x)}{2x} = \frac{5}{2}}$$

Example 2

Evaluate:

$$\lim_{x \rightarrow 0} \frac{\sin(7x)}{14x}$$

Step 1: Rewrite

$$\begin{aligned} &= \frac{7}{14} \cdot \frac{\sin(7x)}{7x} \\ &= \frac{1}{2} \cdot \frac{\sin(7x)}{7x} \end{aligned}$$

Step 2: Use standard limit

$$= \frac{1}{2} \times 1$$
$$= \frac{1}{2}$$

Therefore,

$$\lim_{x \rightarrow 0} \frac{\sin(7x)}{14x} = \frac{1}{2}$$

2. Limit of Type $\frac{\sin ax}{\sin bx}$

Formula

If $a \neq 0$ and $b \neq 0$, then

$$\lim_{x \rightarrow 0} \frac{\sin(ax)}{\sin(bx)} = \frac{a}{b}$$

Example 3

Evaluate:

$$\lim_{x \rightarrow 0} \frac{\sin(4x)}{\sin(9x)}$$

Step 1: Multiply and divide

$$= \frac{\sin(4x)}{4x} \cdot \frac{9x}{\sin(9x)} \cdot \frac{4}{9}$$

Step 2: Apply standard limits

$$\lim_{x \rightarrow 0} \frac{\sin(4x)}{4x} = 1$$

and

$$\lim_{x \rightarrow 0} \frac{9x}{\sin(9x)} = 1$$

Thus,

$$= 1 \times 1 \times \frac{4}{9}$$
$$= \frac{4}{9}$$

Therefore,

$$\lim_{x \rightarrow 0} \frac{\sin(4x)}{\sin(9x)} = \frac{4}{9}$$

Example 4

Evaluate:

$$\lim_{x \rightarrow 0} \frac{\sin(11x)}{\sin(2x)}$$

Using the formula directly:

$$= \frac{11}{2}$$

Hence,

$$\boxed{\lim_{x \rightarrow 0} \frac{\sin(11x)}{\sin(2x)} = \frac{11}{2}}$$

3. Limits Involving Tangent Function**Formula**

Since

$$\lim_{x \rightarrow 0} \frac{\tan x}{x} = 1$$

therefore,

$$\lim_{x \rightarrow 0} \frac{\tan(ax)}{bx} = \frac{a}{b}$$

Example 5

Evaluate:

$$\lim_{x \rightarrow 0} \frac{\tan(3x)}{5x}$$

Rewrite:

$$= \frac{3}{5} \cdot \frac{\tan(3x)}{3x}$$

Using standard limit:

$$= \frac{3}{5} \times 1$$

$$= \frac{3}{5}$$

Therefore,

$$\boxed{\lim_{x \rightarrow 0} \frac{\tan(3x)}{5x} = \frac{3}{5}}$$

Example 6

Evaluate:

$$\lim_{x \rightarrow 0} \frac{\tan(8x)}{\tan(2x)}$$

Using the standard formula:

$$= \frac{8}{2}$$

$$= 4$$

Hence,

$$\boxed{\lim_{x \rightarrow 0} \frac{\tan(8x)}{\tan(2x)} = 4}$$

4. Mixed Trigonometric Limits**Example 7**

Evaluate:

$$\lim_{x \rightarrow 0} \frac{\sin(6x)}{\tan(2x)}$$

Step 1: Rewrite

$$= \frac{\sin(6x)}{6x} \cdot \frac{2x}{\tan(2x)} \cdot \frac{6}{2}$$

Step 2: Apply standard limits

$$= 1 \times 1 \times 3$$

$$= 3$$

Therefore,

$$\boxed{\lim_{x \rightarrow 0} \frac{\sin(6x)}{\tan(2x)} = 3}$$

5. Summary Table

Type of Limit	Result
$\lim_{x \rightarrow 0} \frac{\sin(ax)}{bx}$	$\frac{a}{b}$
$\lim_{x \rightarrow 0} \frac{\sin(ax)}{\sin(bx)}$	$\frac{a}{b}$
$\lim_{x \rightarrow 0} \frac{\tan(ax)}{bx}$	$\frac{a}{b}$
$\lim_{x \rightarrow 0} \frac{\tan(ax)}{\tan(bx)}$	$\frac{a}{b}$
$\lim_{x \rightarrow 0} \frac{\sin(ax)}{\tan(bx)}$	$\frac{a}{b}$

6. Practice Questions

1. Evaluate:

$$\lim_{x \rightarrow 0} \frac{\sin(9x)}{3x}$$

2. Find:

$$\lim_{x \rightarrow 0} \frac{\sin(5x)}{\sin(8x)}$$

3. Evaluate:

$$\lim_{x \rightarrow 0} \frac{\tan(4x)}{7x}$$

4. Find:

$$\lim_{x \rightarrow 0} \frac{\tan(10x)}{\tan(5x)}$$

5. Evaluate:

$$\lim_{x \rightarrow 0} \frac{\sin(12x)}{\tan(3x)}$$

6. Find:

$$\lim_{x \rightarrow 0} \frac{2 \sin(7x)}{x}$$

7. Evaluate:

$$\lim_{x \rightarrow 0} \frac{x}{\sin(6x)}$$

8. Find:

$$\lim_{x \rightarrow 0} \frac{\tan(9x)}{\sin(3x)}$$

Important Note

Whenever limits involve trigonometric functions near $x = 0$, try converting them into the standard forms:

$$\frac{\sin x}{x} \quad \text{or} \quad \frac{\tan x}{x}$$

Then apply the standard limits.

Limits of Piecewise and Absolute Value Functions

Introduction

For piecewise and absolute value functions, we usually evaluate:

- Left-hand limit (LHL)
- Right-hand limit (RHL)
- Compare both limits

If

$$\text{LHL} = \text{RHL}$$

then the limit exists.

1. Limits of Piecewise Functions

Example 1

Find:

$$\lim_{x \rightarrow 2} f(x)$$

where

$$f(x) = \begin{cases} x + 4, & x < 2 \\ 2x + 1, & x \geq 2 \end{cases}$$

Step 1: Find Left-Hand Limit

For $x < 2$,

$$f(x) = x + 4$$

Therefore,

$$\lim_{x \rightarrow 2^-} f(x) = 2 + 4 = 6$$

Step 2: Find Right-Hand Limit

For $x \geq 2$,

$$f(x) = 2x + 1$$

Thus,

$$\lim_{x \rightarrow 2^+} f(x) = 2(2) + 1 = 5$$

Step 3: Compare

Since

$$6 \neq 5$$

therefore,

$\lim_{x \rightarrow 2} f(x)$ does not exist

Example 2

Find:

$$\lim_{x \rightarrow 1} f(x)$$

where

$$f(x) = \begin{cases} 3x - 1, & x < 1 \\ x^2 + 1, & x \geq 1 \end{cases}$$

Left-Hand Limit

$$\lim_{x \rightarrow 1^-} f(x) = 3(1) - 1 = 2$$

Right-Hand Limit

$$\lim_{x \rightarrow 1^+} f(x) = 1^2 + 1 = 2$$

Since both are equal,

$\lim_{x \rightarrow 1} f(x) = 2$

2. Absolute Value Functions

Important Property

The absolute value function is defined as:

$$|x| = \begin{cases} x, & x \geq 0 \\ -x, & x < 0 \end{cases}$$

Example 3

Evaluate:

$$\lim_{x \rightarrow 0} |x|$$

Left-Hand Limit

For $x < 0$,

$$|x| = -x$$

Thus,

$$\lim_{x \rightarrow 0^-} |x| = \lim_{x \rightarrow 0^-} (-x) = 0$$

Right-Hand Limit

For $x > 0$,

$$|x| = x$$

Therefore,

$$\lim_{x \rightarrow 0^+} |x| = \lim_{x \rightarrow 0^+} x = 0$$

Since both limits are equal,

$$\boxed{\lim_{x \rightarrow 0} |x| = 0}$$

Example 4

Evaluate:

$$\lim_{x \rightarrow 0} \frac{|x|}{x}$$

Left-Hand Limit

For $x < 0$,

$$|x| = -x$$

So,

$$\frac{|x|}{x} = \frac{-x}{x} = -1$$

Hence,

$$\lim_{x \rightarrow 0^-} \frac{|x|}{x} = -1$$

Right-Hand Limit

For $x > 0$,

$$|x| = x$$

Thus,

$$\frac{|x|}{x} = \frac{x}{x} = 1$$

Therefore,

$$\lim_{x \rightarrow 0^+} \frac{|x|}{x} = 1$$

Since

$$-1 \neq 1$$

the limit does not exist.

$\lim_{x \rightarrow 0} \frac{ x }{x} \text{ does not exist}$

Example 5

Find:

$$\lim_{x \rightarrow 3} |x - 3|$$

Left-Hand Limit

For $x < 3$,

$$|x - 3| = -(x - 3)$$

$$= 3 - x$$

Thus,

$$\lim_{x \rightarrow 3^-} |x - 3| = 0$$

Right-Hand Limit

For $x > 3$,

$$|x - 3| = x - 3$$

Therefore,

$$\lim_{x \rightarrow 3^+} |x - 3| = 0$$

Since both are equal,

$$\boxed{\lim_{x \rightarrow 3} |x - 3| = 0}$$

3. Finding Unknown Constants

Example 6

Suppose

$$f(x) = \begin{cases} a + x, & x < 2 \\ 5, & x = 2 \\ b - x, & x > 2 \end{cases}$$

If

$$\lim_{x \rightarrow 2} f(x) = f(2)$$

find the values of a and b .

Step 1: Use continuity condition

Since

$$\lim_{x \rightarrow 2} f(x) = f(2) = 5$$

both left-hand and right-hand limits must equal 5.

Step 2: Left-Hand Limit

$$\lim_{x \rightarrow 2^-} f(x) = a + 2$$

Set equal to 5:

$$a + 2 = 5$$

$$a = 3$$

Step 3: Right-Hand Limit

$$\lim_{x \rightarrow 2^+} f(x) = b - 2$$

Set equal to 5:

$$b - 2 = 5$$

$$b = 7$$

Therefore,

$a = 3,$	$b = 7$
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4. Summary Table

Situation	Method
Piecewise Function	Find LHL and RHL separately
Absolute Value Function	Use definition of $ x $
Limit Exists	$\text{LHL} = \text{RHL}$
Limit Does Not Exist	$\text{LHL} \neq \text{RHL}$
Finding Constants	Use continuity condition

5. Practice Questions

1. Find:

$$\lim_{x \rightarrow 1} f(x)$$

where

$$f(x) = \begin{cases} 2x + 1, & x < 1 \\ x^2 + 2, & x \geq 1 \end{cases}$$

2. Evaluate:

$$\lim_{x \rightarrow 0} |2x|$$

3. Find:

$$\lim_{x \rightarrow 0} \frac{|x + 1| - 1}{x}$$

4. Evaluate:

$$\lim_{x \rightarrow 4} |x - 4|$$

5. Determine whether the limit exists:

$$\lim_{x \rightarrow 0} \frac{|x|}{x^2}$$

6. Find a and b if

$$f(x) = \begin{cases} ax + 1, & x < 3 \\ 7, & x = 3 \\ b - x, & x > 3 \end{cases}$$

and

$$\lim_{x \rightarrow 3} f(x) = f(3)$$

Important Note

For piecewise and absolute value functions:

- Always check both sides separately.
- Never assume the limit exists automatically.
- Compare left-hand and right-hand limits carefully.

Indeterminate Forms and Their Solutions

What are Indeterminate Forms?

Sometimes, after directly substituting a value into a limit, we do not get a definite answer.

Instead, we get expressions like:

$$\frac{0}{0}, \quad \frac{\infty}{\infty}, \quad 0 \times \infty$$

These are called **indeterminate forms**.

They are called indeterminate because the actual value of the limit is not immediately known.

1. Common Indeterminate Forms

Indeterminate Form	Name
$\frac{0}{0}$	Zero upon Zero
$\frac{\infty}{\infty}$	Infinity upon Infinity
$0 \times \infty$	Zero times Infinity
$\infty - \infty$	Infinity minus Infinity
0^0	Zero power Zero
∞^0	Infinity power Zero
1^∞	One power Infinity

2. Indeterminate Form $\frac{0}{0}$

Method

If substitution gives

$$\frac{0}{0}$$

then use methods like:

- Factorization
- Rationalization
- Trigonometric identities
- Algebraic simplification

Example 1 : Factorization

Evaluate:

$$\lim_{x \rightarrow 3} \frac{x^2 - 9}{x - 3}$$

Step 1: Substitute

$$= \frac{3^2 - 9}{3 - 3} = \frac{0}{0}$$

This is indeterminate.

Step 2: Factorize

$$x^2 - 9 = (x - 3)(x + 3)$$

Thus,

$$\frac{x^2 - 9}{x - 3} = x + 3$$

Step 3: Substitute again

$$= 3 + 3$$

$$= 6$$

Therefore,

$$\boxed{\lim_{x \rightarrow 3} \frac{x^2 - 9}{x - 3} = 6}$$

3. Rationalization Method

Example 2

Evaluate:

$$\lim_{x \rightarrow 0} \frac{\sqrt{x+4} - 2}{x}$$

Step 1: Substitute

$$= \frac{2 - 2}{0} = \frac{0}{0}$$

Indeterminate form.

Step 2: Rationalize

Multiply numerator and denominator by conjugate:

$$\frac{\sqrt{x+4} - 2}{x} \cdot \frac{\sqrt{x+4} + 2}{\sqrt{x+4} + 2}$$

Using

$$(a - b)(a + b) = a^2 - b^2$$

we get:

$$\begin{aligned} &= \frac{x + 4 - 4}{x(\sqrt{x+4} + 2)} \\ &= \frac{x}{x(\sqrt{x+4} + 2)} \end{aligned}$$

Cancel x :

$$= \frac{1}{\sqrt{x+4}+2}$$

Step 3: Substitute

$$= \frac{1}{2+2}$$
$$= \frac{1}{4}$$

Hence,

$$\boxed{\lim_{x \rightarrow 0} \frac{\sqrt{x+4}-2}{x} = \frac{1}{4}}$$

4. Trigonometric Indeterminate Forms

Important Standard Limits

$$\lim_{x \rightarrow 0} \frac{\sin x}{x} = 1$$

and

$$\lim_{x \rightarrow 0} \frac{\tan x}{x} = 1$$

Example 3

Evaluate:

$$\lim_{x \rightarrow 0} \frac{\sin(5x)}{x}$$

Rewrite:

$$= 5 \cdot \frac{\sin(5x)}{5x}$$

Using standard limit:

$$= 5(1)$$

$$= 5$$

Therefore,

$$\boxed{\lim_{x \rightarrow 0} \frac{\sin(5x)}{x} = 5}$$

5. Indeterminate Form $\infty - \infty$

Method

When a limit gives:

$$\infty - \infty$$

combine terms into a single fraction or rationalize.

Example 4

Evaluate:

$$\lim_{x \rightarrow \infty} (\sqrt{x^2 + x} - x)$$

Step 1: Direct substitution

$$\infty - \infty$$

Indeterminate form.

Step 2: Rationalize

Multiply by conjugate:

$$\begin{aligned} & \frac{\sqrt{x^2 + x} - x}{1} \cdot \frac{\sqrt{x^2 + x} + x}{\sqrt{x^2 + x} + x} \\ &= \frac{x^2 + x - x^2}{\sqrt{x^2 + x} + x} \\ &= \frac{x}{\sqrt{x^2 + x} + x} \end{aligned}$$

Step 3: Divide numerator and denominator by x

$$= \frac{1}{\sqrt{1 + \frac{1}{x}} + 1}$$

As $x \rightarrow \infty$,

$$\frac{1}{x} \rightarrow 0$$

Thus,

$$\begin{aligned} &= \frac{1}{1 + 1} \\ &= \frac{1}{2} \end{aligned}$$

Therefore,

$$\boxed{\lim_{x \rightarrow \infty} (\sqrt{x^2 + x} - x) = \frac{1}{2}}$$

6. Indeterminate Form $\frac{\infty}{\infty}$

Method

For

$$\frac{\infty}{\infty}$$

divide numerator and denominator by the highest power of x .

Example 5

Evaluate:

$$\lim_{x \rightarrow \infty} \frac{3x^2 + 2x + 1}{x^2 + 5}$$

Step 1: Divide by x^2

$$= \frac{3 + \frac{2}{x} + \frac{1}{x^2}}{1 + \frac{5}{x^2}}$$

Step 2: Take limit

As $x \rightarrow \infty$,

$$\frac{2}{x} \rightarrow 0, \quad \frac{1}{x^2} \rightarrow 0$$

Thus,

$$\begin{aligned} &= \frac{3 + 0 + 0}{1 + 0} \\ &= 3 \end{aligned}$$

Hence,

$$\boxed{\lim_{x \rightarrow \infty} \frac{3x^2 + 2x + 1}{x^2 + 5} = 3}$$

7. Summary Table

Indeterminate Form	Method Used
$\frac{0}{0}$	Factorization / Rationalization
$\frac{\infty}{\infty}$	Divide by highest power
$\infty - \infty$	Rationalization
Trigonometric Limits	Use standard identities

8. Practice Questions

1. Evaluate:

$$\lim_{x \rightarrow 2} \frac{x^2 - 4}{x - 2}$$

2. Find:

$$\lim_{x \rightarrow 0} \frac{\sqrt{x+9} - 3}{x}$$

3. Evaluate:

$$\lim_{x \rightarrow 0} \frac{\sin(8x)}{x}$$

4. Find:

$$\lim_{x \rightarrow \infty} \frac{5x^2 + 1}{2x^2 - 3}$$

5. Evaluate:

$$\lim_{x \rightarrow \infty} \left(\sqrt{x^2 + 4x} - x \right)$$

6. Find:

$$\lim_{x \rightarrow 1} \frac{x^3 - 1}{x - 1}$$

7. Evaluate:

$$\lim_{x \rightarrow 0} \frac{1 - \cos x}{x^2}$$

8. Find:

$$\lim_{x \rightarrow \infty} \frac{7x^3 + 2}{x^3 - 5x}$$

Important Note

Whenever direct substitution gives an indeterminate form:

$$\frac{0}{0}, \quad \frac{\infty}{\infty}, \quad \infty - \infty$$

do not stop there.

Simplify the expression first, then evaluate the limit.

Solved Examples

Example 1

Evaluate:

$$\lim_{x \rightarrow 2} (3x + 4)$$

Solution

Substitute $x = 2$:

$$= 3(2) + 4$$

$$= 10$$

$$\boxed{10}$$

Example 2

Evaluate:

$$\lim_{x \rightarrow 1} (x^2 + 5x - 2)$$

Solution

$$= 1^2 + 5(1) - 2$$

$$= 4$$

$$\boxed{4}$$

Example 3

Evaluate:

$$\lim_{x \rightarrow 3} \frac{x + 2}{x + 1}$$

Solution

$$= \frac{3 + 2}{3 + 1}$$

$$= \frac{5}{4}$$

$$\boxed{\frac{5}{4}}$$

Example 4

Evaluate:

$$\lim_{x \rightarrow 4} \frac{x^2 - 16}{x - 4}$$

Solution

Direct substitution gives:

$$\frac{0}{0}$$

Factorize:

$$x^2 - 16 = (x - 4)(x + 4)$$

$$= \frac{(x - 4)(x + 4)}{x - 4}$$

$$= x + 4$$

Substitute $x = 4$:

$$= 8$$

$$\boxed{8}$$

Example 5

Evaluate:

$$\lim_{x \rightarrow 1} \frac{x^3 - 1}{x - 1}$$

Solution

$$x^3 - 1 = (x - 1)(x^2 + x + 1)$$

$$= x^2 + x + 1$$

Substitute $x = 1$:

$$= 1 + 1 + 1$$

$$= 3$$

$$\boxed{3}$$

Example 6

Evaluate:

$$\lim_{x \rightarrow 0} \frac{\sin(6x)}{x}$$

Solution

$$= 6 \cdot \frac{\sin(6x)}{6x}$$

Using

$$\lim_{x \rightarrow 0} \frac{\sin x}{x} = 1$$

$$= 6(1)$$

$$= 6$$

$$\boxed{6}$$

Example 7

Evaluate:

$$\lim_{x \rightarrow 0} \frac{\tan(5x)}{x}$$

Solution

$$= 5 \cdot \frac{\tan(5x)}{5x}$$

Using

$$\lim_{x \rightarrow 0} \frac{\tan x}{x} = 1$$

$$= 5$$

$$\boxed{5}$$

Example 8

Evaluate:

$$\lim_{x \rightarrow 0} \frac{\sin(3x)}{\sin(7x)}$$

Solution

$$= \frac{3}{7}$$

$$\boxed{\frac{3}{7}}$$

Example 9

Evaluate:

$$\lim_{x \rightarrow 0} \frac{\sqrt{x+1} - 1}{x}$$

Solution

Multiply by conjugate:

$$\begin{aligned} &= \frac{\sqrt{x+1} - 1}{x} \cdot \frac{\sqrt{x+1} + 1}{\sqrt{x+1} + 1} \\ &= \frac{x}{x(\sqrt{x+1} + 1)} \\ &= \frac{1}{\sqrt{x+1} + 1} \end{aligned}$$

Substitute $x = 0$:

$$= \frac{1}{2}$$

$$\boxed{\frac{1}{2}}$$

Example 10

Evaluate:

$$\lim_{x \rightarrow \infty} \frac{4x^2 + 2x + 1}{2x^2 + 5}$$

Solution

Divide by x^2 :

$$= \frac{4 + \frac{2}{x} + \frac{1}{x^2}}{2 + \frac{5}{x^2}}$$

As $x \rightarrow \infty$,

$$\frac{2}{x} \rightarrow 0, \quad \frac{1}{x^2} \rightarrow 0$$

$$= \frac{4}{2}$$

$$= 2$$

$$\boxed{2}$$

Example 11

Evaluate:

$$\lim_{x \rightarrow 0} |x|$$

Solution

Both left-hand and right-hand limits equal 0.
Therefore,

$$\boxed{0}$$

Example 12

Evaluate:

$$\lim_{x \rightarrow 0} \frac{|x|}{x}$$

Solution

For $x < 0$:

$$\frac{|x|}{x} = -1$$

For $x > 0$:

$$\frac{|x|}{x} = 1$$

Since LHL $\neq$ RHL,

Limit does not exist

Example 13

Evaluate:

$$\lim_{x \rightarrow 2} f(x)$$

where

$$f(x) = \begin{cases} x + 2, & x < 2 \\ 3x - 1, & x \geq 2 \end{cases}$$

Solution

LHL:

$$2 + 2 = 4$$

RHL:

$$3(2) - 1 = 5$$

Since

$$4 \neq 5$$

Limit does not exist

Example 14

Evaluate:

$$\lim_{x \rightarrow 3} |x - 3|$$

Solution

From both sides, the value approaches 0.

Therefore,

$$\boxed{0}$$

Example 15

Evaluate:

$$\lim_{x \rightarrow \infty} (\sqrt{x^2 + 3x} - x)$$

Solution

Multiply by conjugate:

$$\begin{aligned} &= \frac{x^2 + 3x - x^2}{\sqrt{x^2 + 3x} + x} \\ &= \frac{3x}{\sqrt{x^2 + 3x} + x} \end{aligned}$$

Divide numerator and denominator by x :

$$= \frac{3}{\sqrt{1 + \frac{3}{x}} + 1}$$

As $x \rightarrow \infty$,

$$\begin{aligned} &\frac{3}{1 + 1} \\ &= \frac{3}{2} \end{aligned}$$

$$\boxed{\frac{3}{2}}$$

Exercise

1. Evaluate:

$$\lim_{x \rightarrow 5} (2x - 7)$$

2. Find:

$$\lim_{x \rightarrow 2} (x^2 + 4x + 1)$$

3. Evaluate:

$$\lim_{x \rightarrow 1} \frac{x^2 - 1}{x - 1}$$

4. Find:

$$\lim_{x \rightarrow 0} \frac{\sin(8x)}{x}$$

5. Evaluate:

$$\lim_{x \rightarrow 0} \frac{\tan(7x)}{x}$$

6. Find:

$$\lim_{x \rightarrow 0} \frac{\sin(2x)}{\sin(9x)}$$

7. Evaluate:

$$\lim_{x \rightarrow 0} \frac{\sqrt{x + 16} - 4}{x}$$

8. Find:

$$\lim_{x \rightarrow 0} \frac{|x|}{x}$$

9. Evaluate:

$$\lim_{x \rightarrow \infty} \frac{5x^2 + 3}{x^2 + 2}$$

10. Find:

$$\lim_{x \rightarrow \infty} \left(\sqrt{x^2 + 5x} - x \right)$$

Answer Key

1. 3
2. 13
3. 2
4. 8
5. 7
6. $\frac{2}{9}$
7. $\frac{1}{8}$
8. Does not exist
9. 5
10. $\frac{5}{2}$